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Hops, Yeast, Pure Delight: How Fermentation Temperature Shapes the Aroma of Top-Fermented Beers

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Abstract

Top-fermented beers owe much of their fruity aroma to esters and higher alcohols formed by the yeast during fermentation. This (fictional) thesis investigates how strongly the fermentation temperature shapes this aroma profile. In twelve brews of a pale ale the temperature was varied between 16 °C and 24 °C. Isoamyl acetate – the typical banana note – increased almost linearly by about 38 % per 4 K. A blind tasting with 24 participants confirms: warmer fermented beers are perceived as fruitier, but less “clean”.

Kurzfassung

Obergärige Biere verdanken ihr fruchtiges Aroma vor allem den Estern und höheren Alkoholen, die Hefen während der Gärung bilden. Diese (erfundene) Arbeit untersucht, wie stark die Gärtemperatur dieses Aromaprofil prägt. Isoamylacetat stieg um rund 38 % je 4 K; eine Blindverkostung bestätigt den Effekt.

Acknowledgements

Thanks to everyone who tasted twelve pale ales in one afternoon – all in the name of science.

Contents

1. Introduction	1
1.1. Motivation	1
1.2. Goals and Research Questions	1
1.3. Methodology and Structure	1
2. Background and Related Work	2
2.1. What Yeast Actually Does	2
2.2. Temperature as a Lever	2
2.3. Summary	2
3. Experimental Design	3
3.1. Brews	3
3.2. Analysis and Tasting	3
4. Brewing Trials	4
4.1. Procedure	4
4.2. Sampling	4
5. Results	5
5.1. Ester Formation	5
5.2. Tasting	5
5.3. Discussion	5
6. Lessons Learned	6
7. Future Work	7
8. Conclusion	8
A. Raw Data	10
Bibliography	11
Ludography	12

List of Figures

5.1. Ester concentration depending on fermentation temperature (means, $n = 3$).	5
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List of Tables

3.1. Recipe and fermentation parameters (per 20 L brew)	3
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1. Introduction

Beer is one of the oldest foods of mankind. Since the Bavarian purity law of 1516 it consists of only four ingredients: water, malt, hops and yeast (Hopfner, 2019). Despite this simplicity, beers differ enormously in taste and smell – and a large part of this variety is not created in the brewhouse but in the fermentation tank (Malzbauer & Gerstner, 2021).

1.1. Motivation

Craft breweries increasingly experiment with fermentation parameters. At the same time there is a lack of **systematic data** on how strongly individual parameters actually change the aroma.

1.2. Goals and Research Questions

1. How does the concentration of aroma-active esters change with fermentation temperature?
2. From which temperature do tasters perceive the difference?
3. Can the aroma profile be predicted with a simple model?

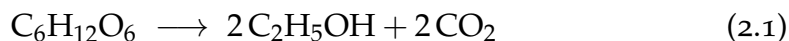
1.3. Methodology and Structure

Chapter 2 explains the biochemistry of fermentation, Chapter 3 the experimental design, Chapter 5 presents measurements and tasting results.

2. Background and Related Work

2.1. What Yeast Actually Does

During fermentation the yeast *Saccharomyces cerevisiae* converts fermentable sugar into ethanol and carbon dioxide:



Along the way hundreds of aroma compounds are formed. The most important ones are

- **esters** such as isoamyl acetate (banana) and ethyl hexanoate (apple),
- **higher alcohols** such as isoamyl alcohol (solvent-like),
- **phenols** such as 4-vinylguaiaicol (clove, typical for wheat beer).

2.2. Temperature as a Lever

Ester formation roughly follows an Arrhenius relation (Gerstner, 2020):

$$k(T) = A \cdot \exp\left(-\frac{E_a}{RT}\right) \quad (2.2)$$

According to Equation 2.2, a higher temperature should accelerate the formation considerably – how strongly, however, is hardly documented for top-fermenting yeasts (Hefeling & Sudmeister, 2018; Malzbauer & Gerstner, 2021).

2.3. Summary

Esters dominate the fruity character of top-fermented beers, and their formation depends on temperature.

3. Experimental Design

3.1. Brews

Twelve identical brews of a pale ale (original gravity 12.5 °P) were split into four temperature levels with three replicates each (Table 3.1).

Table 3.1.: Recipe and fermentation parameters (per 20 L brew)

Ingredient / parameter	Amount	Unit
Pale ale malt	4.2	kg
Cascade hops	60	g
Top-fermenting yeast	11.5	g
Fermentation temperature	16–24	°C
Fermentation time	7	days

3.2. Analysis and Tasting

Esters were measured by gas chromatography. In addition, 24 participants rated the beers blind on fruitiness, cleanness and overall impression (scale 1–9).

4. Brewing Trials

4.1. Procedure

All brews were mashed, boiled and hopped identically on a 50 L pilot plant. After pitching, each fermenter was kept at its target temperature with a deviation of at most 0.5 K.

4.2. Sampling

Samples were taken daily; the final analysis was performed after seven days of fermentation.

5. Results

5.1. Ester Formation

Figure 5.1 shows the clear increase of isoamyl acetate with temperature.

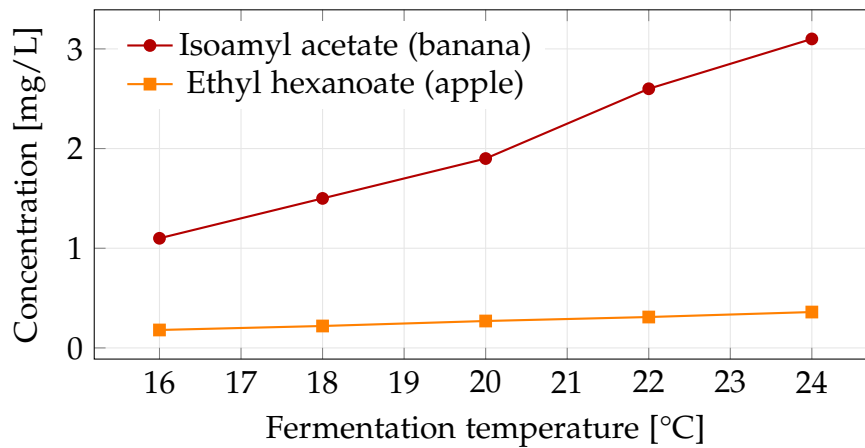


Figure 5.1.: Ester concentration depending on fermentation temperature (means, $n = 3$).

5.2. Tasting

From 20 °C on, 88 % of the participants noticed the difference. The beer fermented at 18 °C received the best overall impression – fruity, but still clean.

5.3. Discussion

The results agree with the kinetic expectation from Equation 2.2.

6. Lessons Learned

Temperature control matters more than any single hop addition – and tasting panels should not start after lunch. Simulations such as *Brewmaster Tycoon* (Fictional Games, 2024) are no substitute for a real brewhouse.

7. Future Work

- Further yeast strains, in particular kveik yeasts,
- temperature profiles instead of constant temperatures,
- and of course: a larger tasting panel.

8. Conclusion

Fermentation temperature is the most effective and at the same time cheapest lever for the aroma of top-fermented beers. Just 4 K change the ester profile more than doubling the aroma hop addition (Hopfner, 2019).

Appendix

Appendix A.

Raw Data

All data of this fictional thesis are made up.

Bibliography

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Ludography

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